

MD MAHFUZUR RAHMAN

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PROFESSIONAL SUMMARY

- Biomedical-focused Mechanical Engineer with 6+ years of academic research experience in developing, testing and validating affordable, wearable sensing systems for noninvasive cardiovascular health assessment — bridging the gap between laboratory-grade clinical instruments and accessible home-use devices. Conducting NSF-funded research (Grant #1936005) with 6 peer-reviewed publications (4 as first author).
- Experienced in end-to-end device prototyping — microfluidic tactile sensors and accelerometer-based handheld and wearable pulse acquisition systems — encompassing microfabrication, custom analog front-end circuitry (transimpedance amplifier, demodulation stage), and full sensor-to-DAQ system integration.
- Expertise in cardiovascular biomechanical modeling and advanced signal-processing algorithms to study arterial pulse dynamics and motion artifacts, integrating digital filter design (FIR: Kaiser window-based, equiripple/Parks-McClellan; IIR: Butterworth; DWT-based; adaptive), FFT, EMD, CDM, HHT, and HVD for time-frequency analysis of nonstationary physiological signals.
- Collaborated across academic and clinical settings on multiple IRB-approved human-subject studies — including NSF-funded projects, clinical research with Sentara Health Research Center, and joint research with the Department of Kinesiology and Health Sciences, University of Waterloo (Canada) — supporting protocol development, physiological data acquisition, and noninvasive cardiovascular health assessment using developed sensing systems.

EDUCATION

Ph.D., Mechanical & Aerospace Engineering

August 2026

Old Dominion University, Norfolk, VA | GPA: 3.87/4.0

- **Dissertation:** Time-Frequency Analysis of Arterial Pulse Signals of Cardiovascular Disease Patients Measured by a Microfluidic Based Tactile Sensor
- **Advisor:** Dr. Zhili Hao | NSF Grant #1936005

M.S., Applied Physics, Electronics & Communication Engineering

March 2012

University of Dhaka, Bangladesh | GPA: 3.73/4.0 (WES)

- **Thesis:** Design of an all-optical switching structure using coupled 2D photonic crystal waveguides for optical communication
- **Advisor:** Dr. Subrata Das

B.S., Applied Physics, Electronics & Communication Engineering

September 2010

University of Dhaka, Bangladesh | GPA: 3.81/4.0 (WES)

- **Project:** Design of a Low-loss Y-Splitter for Optical Telecommunication using a 2D Photonics Crystal
- **Advisor:** Dr. Saeed Mahmud Ullah

PUBLICATIONS

Preprints

2026

- **Rahman, M.M.**, Hasan, M., May, J. F., Herre, J. M., Reynolds, L. J., & Hao, Z. (2026). Subject-Specific Cardiovascular Signatures Revealed by Arterial Pulse Signals in Patients with Heart Disease Part I: General Cases. [doi:10.20944/preprints202608.2302.v1](https://doi.org/10.20944/preprints202608.2302.v1)
- Hasan, M., **Rahman, M.M.**, May, J. F., Herre, J. M., Reynolds, L., & Hao, Z. (2026). Subject-Specific Cardiovascular Signatures Revealed by Arterial Pulse Signals in Patients With Heart Diseases Part II: Special

Journal Articles (Peer-Reviewed)

2026

- **Rahman, M.M.**, Hasan, M., May, J.F., Herre, J.M., Reynolds, L., & Hao, Z. (in press). Radial and carotid arterial pulse signals for assessing cardiovascular function at rest and during post-exercise recovery in a heart transplant patient: a case study. **Medical Sensors & Imaging** (IOP Publishing) [Accepted; [doi:10.1088/2977-8425/ae4b40](https://doi.org/10.1088/2977-8425/ae4b40)]

2025

- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part I: Accelerometers and PPG Sensors. **Sensors**, 25(18), 5710. [doi:10.3390/s25185710](https://doi.org/10.3390/s25185710)
- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part II: Tactile Sensors. **Sensors**, 25(18), 5700. [doi:10.3390/s25185700](https://doi.org/10.3390/s25185700)
- **Rahman, M.M.**, et al. (2025). Motion Artifacts (MA) At-Rest in Measured Arterial Pulse Signals: Time-Varying Amplitude in Each Harmonic and Non-Flat Harmonic-MA-Coupled Baseline. **Biosensors**, 15(9), 578. [doi:10.3390/bios15090578](https://doi.org/10.3390/bios15090578)
- Toraskar, S., **Rahman, M.M.**, & Hao, Z. (2026). Pulse Measurement Alters the True Pulse Signal in an Artery: A Coupled String-SDOF Model. **ASME J. Engineering and Science in Medical Diagnostics and Therapy**, 9(1), 011207. [doi:10.1115/1.4069099](https://doi.org/10.1115/1.4069099)

2023

- Hao, Z., **Rahman, M.M.**, et al. (2023). Axial Wall Displacement at the Common Carotid Artery is Associated With the Lamb Waves. **ASME J. Engineering and Science in Medical Diagnostics and Therapy**, 6(1), 011008. [doi:10.1115/1.4056267](https://doi.org/10.1115/1.4056267)

Conference Proceedings

2021

- **Rahman, M.M.**, et al. (2021). Measurement of Post-Exercise Response of Local Arterial Parameters Using an Adjustable Microfluidic Tactile Sensor. **IEEE EMBC 2021**, pp.1284–1287. [doi:10.1109/EMBC46164.2021.9630432](https://doi.org/10.1109/EMBC46164.2021.9630432)

2020

- **Rahman, M.M.**, et al. (2020). Improved Vibration-Model-Based Analysis for Estimation of Arterial Parameters From Noninvasively Measured Arterial Pulse Signals. **ASME IMECE 2020**, V005T05A079. [doi:10.1115/IMECE2020-24551](https://doi.org/10.1115/IMECE2020-24551)

RESEARCH EXPERIENCE

Graduate Research Assistant — NSF Grant #1936005

Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Sept 2019 – May 2026

Project 1: Soft Polymer Microfluidic Tactile Sensor — Fabrication and Characterization

- Designed and fabricated end-to-end soft polymer based (PDMS) microfluidic tactile sensors for noninvasive arterial pulse acquisition, encompassing photolithography-based Au/Cr electrode patterning on Pyrex glass substrate, SU-8 based microchannel mold fabrication (100–150 μm), PDMS based microstructure casting and plasma bonding with substrate — achieving 1.5 mm spatial transducer spacing and 0.2 μm displacement resolution in static characterization
- Designed PCBs with KiCAD for analog front-end circuit integrating transimpedance amplifiers, demodulator, adder, multiplier, etc., enabling simultaneous multi-transducer readout with NI DAQ and real-time LabVIEW monitoring
- Adopted 3D-print based microchannel molds as a cost- and time-effective alternative to SU-8 based fabrication, reducing cycle time by eliminating spin-coating, UV exposure, and baking steps while maintaining geometry

fidelity and improving surface uniformity

- Characterized full sensor performance — sensitivity, linearity, frequency response, and repeatability — under controlled bench conditions; results validated in clinical pulse acquisition studies (Sensors 2025, Parts I & II)

Project 2: Noninvasive Physiological Measurement — Arterial Stiffness & Motion Artifacts

- Developed an analytical model of motion artifacts (MAs) to mathematically relate baseline drift and a newly defined time-varying system parameters (TVSPs) induced distortion to a measured pulse signal. (Biosensors 2025; Sensors 2025)
- Developed MATLAB and Python signal processing pipelines integrating digital filter design (FIR/IIR, DWT-based, adaptive), FFT, EMD, CDM, HHT, and HVD for time-frequency analysis of nonstationary arterial pulse signals, enabling beat-by-beat instantaneous parameter extraction
- Conducted IRB-approved human subject studies with healthy participants and cardiac rehabilitation patients (Sentara Heart Hospital), validating sensor-derived arterial parameters including pulse wave velocity, augmentation index, and arterial elasticity against clinical reference measurements
- Extracted and validated arterial mechanical parameters (elasticity, viscosity, radius, PWV) using vibration-model-based analysis (IEEE EMBC 2021; ASME IMECE 2020)

Project 3: Comprehensive Arterial Health Assessment Across BMI Groups (IRB #1452174-27) — Ongoing

- Designed and executed a multi-visit IRB-approved clinical study evaluating wearable sensor performance across two BMI groups (<25 and $30\text{--}40\text{ kg/m}^2$) at radial and carotid arteries, investigating how body composition affects arterial pulse signal quality and derived physiological parameters
- Acquired multimodal cardiovascular data at rest and six post-exercise time points (0, 10, 15, 20, 30, 60 min) simultaneously from microfluidic tactile sensors, Doppler ultrasound, applanation tonometry, and brachial blood pressure — enabling direct comparison of wearable-derived and clinically established parameters
- Evaluated two sensor designs differing in PDMS thickness for pulse waveform reproducibility and validated derived indices (heart rate, respiration rate, augmentation index, PWV) against clinical reference standards
- Established full study infrastructure including IRB protocol, REDCap data management, and multi-visit coordination; study temporarily paused pending resumption

Project 4: Feasibility Study — Accelerometer-Based Carotid Arterial Pulse Acquisition (IRB #1744951-11)

- Designed a custom flexible PCB integrating ST Microelectronics analog (LIS344ALH) and digital (LIS2DW12) accelerometers for capturing carotid arterial wall motion
- Developed analog front-end circuitry for DC offset removal and signal amplification, integrated with NI DAQ and real-time LabVIEW monitoring, enabling physiological parameter extraction (heart rate, PWV) from carotid pulse waveforms
- Conducted IRB-approved single-visit feasibility study comparing analog vs. digital accelerometer performance across four anatomical sites per participant (3 repeated 1-min recordings per site), with simultaneous brachial blood pressure and heart rate reference measurements
- Identified fundamental sensing limitations — tape-fixation motion coupling, low SNR at neck surface, and DC offset instability — that directly informed the analytical motion artifact framework subsequently published in Sensors 2025 Part I

TECHNICAL SKILLS

- **Microfabrication:** Photolithography (spin coating, UV exposure, development); Au/Cr thin-film deposition (sputtering); lift-off; plasma treatment; soft lithography (SU-8 based micro-channel mold fabrication)
- **Sensor Design & Electronics:** AutoCAD (electrodes and microchannel layout); Autodesk Inventor (3D mold design for microchannel); Multisim (circuit simulation); KiCad (PCB design for analog front-end circuits: transimpedance amplifier, multiplier, low-pass filter, adder, subtractor); LabVIEW-based real-time DAQ and monitoring
- **Signal Processing & Time-Series Analysis:**
 - **Spectral & Time-Frequency Analysis:** Fast Fourier Transform (FFT); Continuous Wavelet Transform (CWT); Complex Demodulation (CDM)

- **Filter Design & Denoising:** FIR (Kaiser Window, Parks-McClellan/Equiripple); IIR (Butterworth); Adaptive Filtering; Discrete Wavelet Transform (DWT, Multilevel Decomposition)
- **Adaptive Decomposition:** Empirical Mode Decomposition (EMD); Hilbert-Huang Transform (HHT); Hilbert Vibrational Decomposition (HVD)
- **Developed/Novel Methods (Ph.D. Research):** Hybrid masked based EMD-HVD, and hybrid CDM-HVD framework for instantaneous parameter estimation in nonstationary physiological signals
- **Programming & Computation:** MATLAB (primary platform for algorithm implementation, signal processing, data analysis, and numerical computing including ODE45-based differential equation solving); Python (selected implementations and data analysis pipelines); Simulink (system modeling); LabVIEW (real-time data acquisition, visualization, and recording); SPSS (T-tests, ANOVA)
- **Clinical & Human Subjects Research:** IRB protocol design and compliance; REDCap (data collection, entry, and quality management); pulse signal acquisition at radial and carotid arteries; cardiac rehabilitation patient studies (Sentara Health Research Center); multimodal physiological reference measurement (Doppler ultrasound, applanation tonometry, brachial blood pressure) for clinical validation of wearable sensor-derived parameters
- **Cross-Functional Collaboration & Translational Research:** Partnered with clinical and academic collaborators at Sentara Health Research Center and the University of Waterloo (Canada) on NSF-funded, IRB-approved human subject studies; contributed to patient recruitment and screening, protocol design, physiological data acquisition, and experimental documentation supporting noninvasive cardiovascular health assessment

TEACHING EXPERIENCE

Graduate Teaching Assistant — Mechanical & Aerospace Engineering
Old Dominion University, Norfolk, VA

Sept 2019 – Dec 2025

- Served as primary instructor for MAE 203: Material Science Lab — independently designed and delivered laboratory instruction, developed assessments, and managed course delivery across multiple sections
- Delivered laboratory instruction, graded assignments, and mentored undergraduate students across core mechanical engineering courses including dynamics, solid mechanics, materials science, and engineering design

Lecturer(Full Time) — Electronics & Telecommunication Engineering
Atish Dipankar University of Science & Technology, Dhaka, Bangladesh

Oct 2014 – Mar 2017

- Independently developed and taught 14 undergraduate courses in electronics, circuits, instrumentation, digital signal processing, and microprocessors

Lecturer(Part-Time) — Computer Science and Engineering (CSE)
Daffodil Institute of Information Technology, Dhaka, Bangladesh

Aug 2016 – Mar 2017

- Instructed electrical engineering and microprocessor courses

GRANTS & FUNDING

- NSF Grant #1936005 — Noninvasive Cardiovascular Sensing Systems | PI: Dr. Zhili Hao, ODU | Role: Graduate Research Assistant | 2019–Present

HONORS & AWARDS

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| • Tiwari Endowed Graduate Scholarship, Old Dominion University | 2021 |
| • National Science & ICT (NSICT) Fellowship, Bangladesh | 2010 |
| • Merit-Based Scholarship, University of Dhaka | 2008 |

CERTIFICATIONS

- Biomedical Research — Basic, CITI Program
Issued March 2026 · Expires March 2029 · Credential ID: 72079387 (**Active**)

- Biomedical Research — Refresher 1, CITI Program
Issued November 2021 · Expired November 2023 · Credential ID: 43904237
- Biomedical Research — Refresher 2, CITI Program
Issued December 2023 · Expired December 2025 · Credential ID: 57666724
- Biomedical Responsible Conduct of Research, CITI Program
Issued October 2020 · Credential ID: 39136829

PROFESSIONAL & COMMUNITY SERVICE

- **Conference Reviewer:** International Conference on Imaging, Vision & Pattern Recognition (IVPR)
- **Volunteer:** Virginia Medical Reserve Corps (MRC) May 2020–Present
Volunteered in community public-health initiatives including COVID-19 vaccination clinics, community-based health screening programs, patient intake, and health event coordination

INDUSTRY EXPERIENCE

Manager — Merchandising & Production
Afiya Knitwear Ltd., Dhaka, Bangladesh

Apr 2017 – Aug 2019

- Managed end-to-end production planning, costing, quality compliance, and vendor coordination for international apparel manufacturing contracts
- Developed cross-functional project management and operational efficiency skills in multi-stakeholder, deadline-driven environments

REFERENCES

Available upon request.